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- For earth, $\mu_E := 3.986004418 \times 10^{14}$ m^3/sec^2 (from WGS84)

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- Energy is also conserved (if we ignore internal, heat-generating forces)

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- General Relativity

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- Other minor effects: thermal, magnetic, ...

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- $\dot{X}(t) = \dot{X}^*(t) + \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^* [X(t) - X^*(t)] + \cdots,$

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- Or, equivalently

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- Then the equation of motion becomes: $\dot{x}(t) = Ax(t)$, where $A(t) := \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^*$
- The state transition matrix Φ is associated with the homgeneous solution of the linearized differential equation above, and enables future state to be computed from the current state :

$$x(t) = \Phi(t, t_0)x(t_0)$$

and also itself satisfies the matrix differential equation

$$\dot{\Phi}(t, t_0) = A(t, t_0)\Phi(t, t_0),$$

with initial condition $\Phi(t_0, t_0) = I$